

SANGJU, Korea, Republic of

WMO: 471370

Lat: 36.4083N

Lon: 128.1575E

Elev: 98

StdP: 100.16

Time Zone: 9.00 (E09)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-8.0	-21.9	0.5	-6.0	-19.9	0.6	-4.8	9.3	-2.3	7.9	-1.3	2.8	270	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	33.2	24.8	31.8	24.2	30.4	23.2	26.1	30.6	25.4	29.7	24.8	28.8	1.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.1	28.5	24.2	19.3	27.9	23.6	18.7	27.2	81.8	30.9	78.8	29.8	76.1	29.0	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	5.8	4.9	-12.3	34.9	2.4	1.2	-14.1	35.8	-15.5	36.5	-16.9	37.2	-18.7	38.1		
			-13.9	27.0	2.2	0.8	-15.5	27.5	-16.7	28.0	-18.0	28.5	-19.6	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	-0.8	1.6	6.9	12.9	18.6	22.6	25.3	25.6	20.6	14.5	7.6	0.8	(6)
(7)		DBStd	9.81	3.46	3.63	3.86	3.54	2.77	2.30	2.65	2.80	2.71	3.17	3.90	3.61	(7)
(8)		HDD10.0	1076	335	235	112	13	0	0	0	0	4	93	284		(8)
(9)		HDD18.3	2611	593	468	355	164	30	2	0	9	124	323	542		(9)
(10)		CDD10.0	2202	0	0	15	101	267	377	473	484	319	144	21	0	(10)
(11)		CDD18.3	696	0	0	0	3	39	129	215	226	78	6	0	0	(11)
(12)		CDH23.3	5556	0	0	1	59	379	941	1788	1976	391	20	0	0	(12)
(13)		CDH26.7	1770	0	0	0	8	73	260	621	742	64	1	0	0	(13)
(14)	Wind	WSAvg	1.8	2.5	2.3	2.5	2.3	1.8	1.3	1.2	1.1	1.0	1.2	1.8	2.4	(14)
(15)	Precipitation	PrecAvg	1200	21	30	50	89	91	122	285	261	138	47	41	25	(15)
(16)		PrecMax	1852	73	71	91	186	194	262	642	527	379	147	118	44	(16)
(17)		PrecMin	781	2	2	8	28	22	27	101	63	32	3	5	7	(17)
(18)		PrecStd	256	18	19	24	47	45	68	146	139	94	39	30	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	14.3	21.8	27.9	30.7	32.5	34.7	35.5	30.4	25.4	20.7	11.9	(19)
(20)		MCWB	5.0	7.8	11.9	15.5	17.1	20.3	25.3	25.0	23.8	18.0	14.7	7.7		(20)
(21)		2%	DB	7.8	11.7	18.6	24.9	28.6	30.7	33.0	33.6	28.4	23.5	17.9	9.8	(21)
(22)		MCWB	3.2	5.9	9.7	13.6	16.7	20.1	25.1	25.2	21.6	17.1	12.5	5.7		(22)
(23)		5%	DB	5.9	9.7	16.2	22.9	27.1	29.5	31.5	32.2	27.1	21.9	15.9	8.1	(23)
(24)		MCWB	2.0	4.4	8.2	12.5	16.3	20.0	24.7	25.0	20.7	16.0	11.6	4.6		(24)
(25)		10%	DB	4.4	7.7	14.0	20.7	25.6	28.1	30.1	30.8	25.7	20.3	14.0	6.6	(25)
(26)		MCWB	0.8	3.1	7.3	11.4	15.8	19.7	24.1	24.6	20.1	15.1	10.0	3.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	10.5	13.3	18.0	21.3	24.5	26.7	27.1	25.0	20.5	15.9	9.0	(27)
(28)		MCDWB	8.6	12.0	18.9	25.2	25.1	28.1	31.3	32.4	28.9	22.8	19.2	11.3		(28)
(29)		2%	WB	3.9	6.6	11.4	15.5	19.2	23.0	26.1	26.3	23.7	18.9	13.8	6.5	(29)
(30)		MCDWB	6.5	10.4	15.8	21.8	24.9	26.9	30.4	31.1	26.3	21.9	16.7	8.6		(30)
(31)		5%	WB	2.5	4.9	9.7	13.9	18.0	22.1	25.4	25.7	22.6	17.3	12.2	5.0	(31)
(32)		MCDWB	5.0	8.3	15.0	19.9	23.9	26.2	29.6	30.1	24.9	20.3	15.2	7.5		(32)
(33)		10%	WB	1.3	3.6	8.1	12.5	17.1	21.2	24.8	25.0	21.4	15.8	10.6	3.6	(33)
(34)		MCDWB	4.0	7.1	13.1	18.3	22.7	25.5	28.9	29.2	24.2	19.5	13.5	6.4		(34)
(35)	Mean Daily Temperature Range		MDBR	8.1	9.2	10.6	11.8	11.6	9.7	7.0	7.6	8.9	11.1	9.6	8.1	(35)
(36)		5% DB	MCDBR	9.8	12.5	14.4	16.2	14.8	12.4	9.2	10.0	10.6	12.4	12.0	10.0	(36)
(37)		MCWBR	6.6	8.0	7.7	7.1	5.9	4.3	3.2	3.5	4.7	6.3	7.2	7.0	(37)	
(38)		5% WB	MCDBR	9.0	11.3	12.6	13.1	11.3	8.8	8.0	8.5	7.8	9.8	10.1	9.3	(38)
(39)		MCWBR	6.6	8.0	7.8	6.8	5.7	3.9	3.3	3.7	4.1	5.7	6.8	7.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.458	0.542	0.575	0.585	0.644	0.600	0.551	0.479	0.457	0.435	0.403	(40)	
(41)		taud	2.014	1.927	1.764	1.741	1.763	1.701	1.867	1.988	2.132	2.062	2.041	2.050	(41)	
(42)		Ebn at Noon	741	752	723	729	730	686	714	740	776	747	708	716	(42)	
(43)		Edh at Noon	132	163	210	226	224	238	201	174	143	140	128	120	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.33	4.32	4.96	5.59	5.12	4.31	4.36	3.94	3.58	2.60	2.32	(44)	
(45)		RadStd	0.15	0.30	0.42	0.39	0.50	0.40	0.57	0.59	0.45	0.30	0.32	0.17	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+118
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+77	+42

Nomenclature: See separate page